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“Beyond the Headline: Forecasting Philippine CPI Components and Inflation Drivers”

Abstract:

Monthly CPI data for the Philippines from 2018 to 2025 are used to forecast overall inflation and its component categories. We apply several forecasting methods, both statistical (ARIMA and SARIMA) and neural network models (LSTM, RNN, DNN and CNN), and we use time series cross validation to measure accuracy. All models point to modest inflation over the next year, with CPI rising by roughly 2 to 4 percent by late 2026, and they consistently identify Food, Transport and Financial Services as leading contributors. On a normalized error scale the recurrent neural network (RNN) gives the lowest average RMSE of about 0.049, while ARIMA and SARIMA are slightly higher around 0.075 to 0.078, but the differences are small and not statistically significant. Because SARIMA is accurate, interpretable and easy to apply across all CPI categories, it is adopted as the main forecasting engine and the neural networks are used as benchmarks. The results suggest that policy should focus on stabilizing food and transport costs in order to keep inflation close to the three percent target.

Executive Summary:

This executive summary presents the main results of a forecasting study on Philippine consumer prices. Using monthly CPI data from 2018 to 2025, the study projects headline CPI and major CPI categories twelve months into the future. The focus is on what the forecasts say about future inflation levels and which categories are

expected to rise the fastest. This section highlights the key findings and practical recommendations.

Key Findings: The forecast shows Philippine CPI (All Items) rising moderately by late 2026. ARIMA and SARIMA models predict CPI around 132 to 133 index points by October 2026, up from about 128.6 in late 2025, which implies roughly three percent inflation over twelve months. The neural network models give very similar ranges for the same horizon. CNN forecasts about 130.7, while DNN forecasts about 133.8. In cross validation on a normalized scale the RNN model has the lowest average RMSE (about 0.049), followed by DNN and CNN. ARIMA and SARIMA have slightly higher errors (about 0.075 to 0.078), but all models are in the same narrow band of accuracy. The largest forecast percentage increases over the next year occur in Financial Services (about plus 4.9 percent), Food and non alcoholic beverages including Food itself (about plus 4.3 to 4.6 percent), and Transport (about plus 4.1 percent). These categories are therefore the main inflation drivers in the model.

Key Recommendations: Use the CPI forecasts to guide fiscal planning and wage adjustments

1. **Strengthen local food production** by investing in irrigation, farm machines, post-harvest facilities, cold storage and farm-to-market roads, and by giving small farmers cheaper credit, insurance and training.
2. **Use flexible, predictable import and tariff policies** so rice and other staples remain affordable without suddenly hurting local producers.

3. **Maintain and upgrade buffer stocks** for key goods such as rice, supported by better storage and logistics, so government can release stocks when retail prices spike.
4. **Reduce long-term dependence on imported fuel** by encouraging local renewable energy and more efficient public transport.

Households can plan for slightly higher prices on food and transport by setting aside a small buffer in their monthly budget, prioritising basic needs, and avoiding unnecessary debt. Workers and unions can use the projected three percent inflation as a reference point when discussing wage adjustments. Families who are able to save can build or increase an emergency fund so they are less exposed if actual inflation turns out higher than forecast.

Introduction

The Philippine economy, like any other, is strongly affected by changes in the cost of living, yet most public discussion still focuses on past inflation rather than on what is likely to happen next. The Consumer Price Index or CPI is the main official measure of inflation and is used to guide wage agreements, social protection programs and monetary policy decisions. However, headline CPI alone does not reveal which parts of the household basket are driving price pressure or where future risks may come from. This study uses monthly CPI data for the Philippines from 2018 to 2025 to build a forward looking view of inflation over the next twelve months. It combines traditional time series models, such as ARIMA and SARIMA, with more recent neural network models, and evaluates them through time series cross validation. The goal is not only to

forecast the overall CPI path but also to identify which CPI components, such as food, transport and financial services, are expected to contribute most to future inflation so that policymakers, firms and households can plan ahead more effectively.

Background and Problem Statement:

The Philippines aims for low, stable inflation (target $3.0\% \pm 1.0\%$ for 2025–28). In practice, monthly CPI is volatile: some categories (food, fuel, services) can surge, driving inflation above target. Accurate short-term forecasting is crucial for policy and business planning. Traditional forecasts often focus on headline CPI without examining components. Our problem is to forecast the overall CPI and identify which CPI categories will push inflation up or down. A reliable method can help decision-makers anticipate inflation pressures rather than reacting after the fact.

Objectives, Scope and Significance:

This analysis aims to predict Philippine CPI (All Items) one year ahead and to forecast major CPI components, thereby identifying key inflation drivers. Specifically, we forecast 12 months ahead for the headline CPI and each major category, and compare multiple modeling approaches. We focus on post-2018 monthly CPI data for 15 categories (All Items and 14 sub-groups). The results help gauge whether inflation is likely to stay within the 3% target range, and pinpoint which sectors need policy attention. By revealing category-level drivers, the study informs targeted interventions (e.g. food supply measures) and improves budgeting.

Data Description:

The dataset (“CPI-Many.xlsx”) contains Philippine CPI indices (base 2018=100) for All Items and 14 categories from Jan 2018 through Oct 2025. We extracted the “0 – ALL ITEMS” series to create a monthly time series of CPI, resulting in 94 observations (2018-01 to 2025-10). Similarly, we loaded all available categories under “PHILIPPINES” (forward-filling blank rows) and kept series with ≥ 24 values. In total we obtained 15 series: All Items, Food & Non-Alc Beverages, Alcohol/Tobacco, Clothing, Housing, Furnishings, Health, Transport, Information/Communication, Recreation/Culture, Education, Restaurants/Accom, Financial Services, Personal Care & Miscellaneous, etc. (e.g. ‘01 – FOOD’, ‘07 – TRANSPORT’, ‘12 – FINANCIAL SERVICES’). Missing values (marked ‘.’ in raw data) were omitted. The All-Items CPI has gradually risen from ~97 in early 2018 to ~128.6 in Oct 2025. This range and the number of observations are typical of official CPI series.

Methodology:

We compared statistical and machine-learning time-series models. First, we used ARIMA models: a nonseasonal ARIMA(2,1,2) and a seasonal SARIMA(2,1,2)(1,1,1,12), chosen after testing for best fit. These capture trends, shocks, and yearly seasonality. Second, we built deep learning models: LSTM, SimpleRNN, DNN, and 1D-CNN neural networks, each trained on past CPI data. All models used a “walk-forward” cross-validation: the 94-month series was split into 5 sequential folds, training on early data and testing on later points. The forecast error (RMSE) was computed for each fold, then averaged. This approach mimics real-time forecasting. We evaluated accuracy with RMSE: lower RMSE means better predictions. Finally, each model was trained on the full series to produce a 12-month forecast (Nov 2025 – Oct 2026).

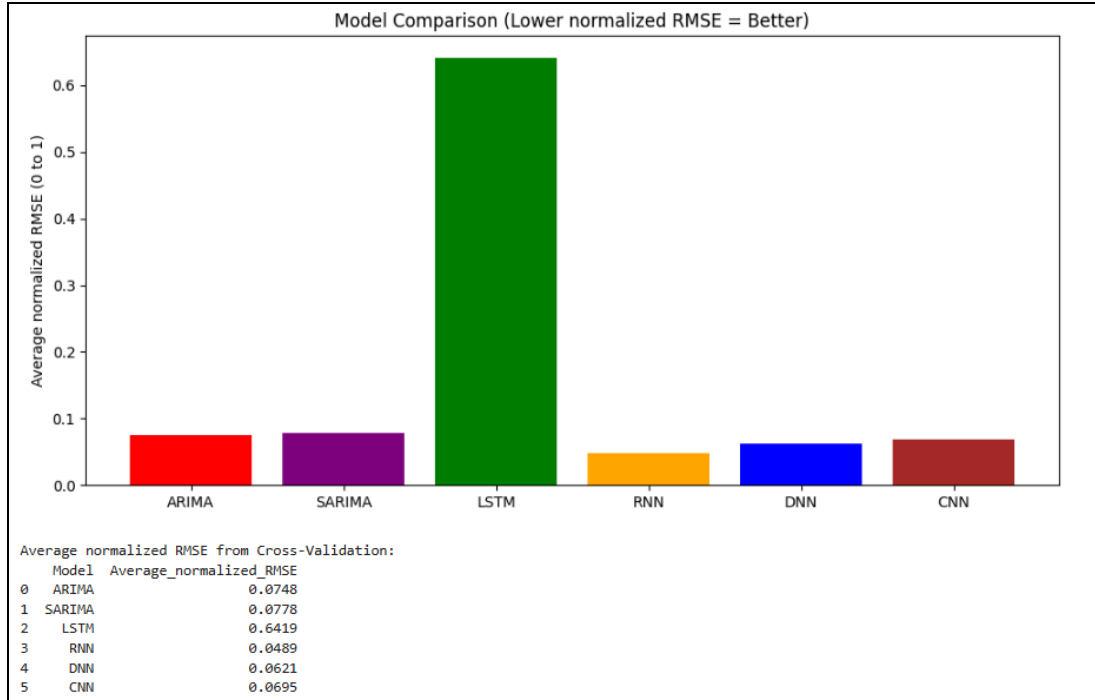


Figure 1: Model Comparison

Figure 1 shows the average normalized RMSE for all models. The RNN achieves the lowest error (about 0.049), followed by DNN and CNN. ARIMA and SARIMA have slightly higher errors (around 0.075 to 0.078), while LSTM performs poorly. Although some neural network models achieve lower error than SARIMA, the differences are small. Given the limited sample size and the need for an interpretable and stable model, SARIMA(2,1,2)(1,1,1,12) is used as the main forecasting model, with neural networks serving as benchmarks.

The raw CPI series is trending upward over time. This means it is non-stationary and violates the basic assumption of ARIMA-type models. To address this, we worked with a transformed version of CPI that is stationary.

First, we took a regular first difference, which is the month-to-month change in CPI. Second, we applied a seasonal difference with lag 12, which subtracts the value from the same month a year earlier. The resulting series is $\Delta\Delta_{12}\text{CPI}$. Statistical tests and plots showed that this differenced series fluctuates around zero with roughly constant variance, so it is close to stationary.

The SARIMA model is fitted and forecasted on this stationary series. For each step ahead, SARIMA produces forecasts of $\Delta\Delta_{12}\text{CPI}$. To obtain the final CPI forecasts shown in the results, these forecasts are then transformed back to levels. This is done by reversing the differencing operations: the predicted differences are added back cumulatively to the last observed CPI values, restoring the original CPI index scale.”

This makes clear that the model works on differences but you present forecasts in the original units.

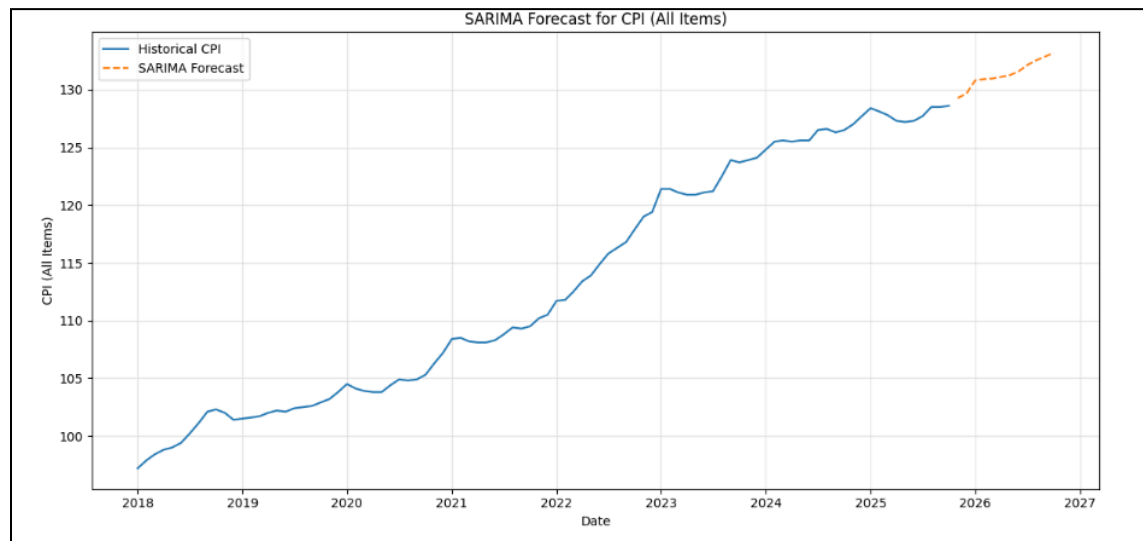


Figure 2: SARIMA Forecast for CPI All Items

Figure 2 shows the SARIMA forecast for headline CPI in its original index form. The blue line is the historical CPI from 2018 to 2025. The orange dashed line is the twelve month forecast obtained after converting the model's differenced predictions back into CPI levels. The forecast continues the upward trend, with CPI expected to rise from about 128.6 in late 2025 to roughly 132–133 by late 2026. This corresponds to moderate inflation rather than a sudden jump in prices.

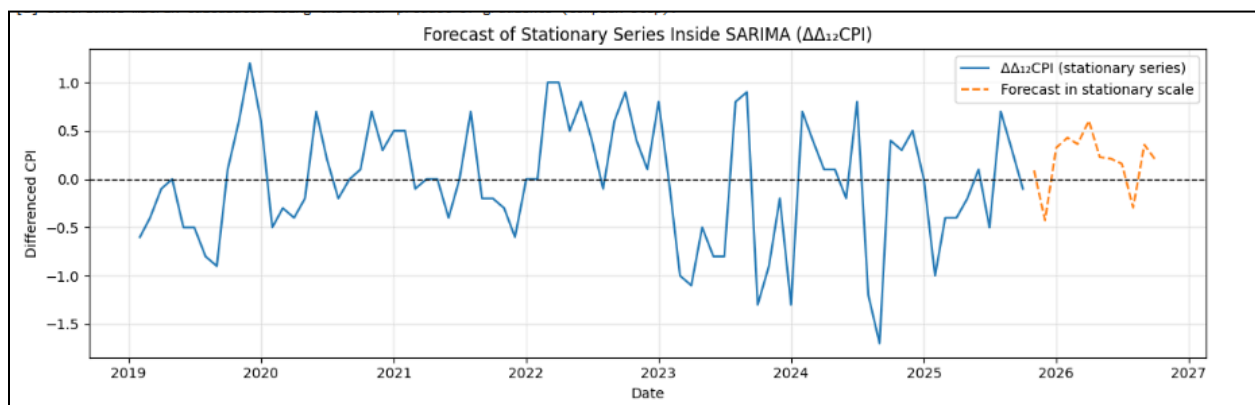


Figure 3: Forecast of Stationary Series inside SARIMA

Figure 3 shows what the SARIMA model actually 'sees' internally. The blue line is the stationary series $\Delta\Delta_{12}CPI$, which represents combined first and seasonal differences of CPI. This series oscillates around zero rather than trending upward, which satisfies the stationarity requirement of the model. The orange dashed line is the twelve month forecast of $\Delta\Delta_{12}CPI$. The predicted values also fluctuate around zero, indicating that the model expects future monthly and seasonal changes to be similar to recent history. When these predicted differences are added back to the last observed CPI values, they produce the level forecasts shown earlier in Figure 2.

Results – Headline CPI Forecast: All models predict modest inflation. The table below (excerpted) shows the 12-month forecast (Nov 2025–Oct 2026) for All-Items CPI from each model

Date	ARIMA	SARIMA	LSTM	RNN	DNN	CNN
2026-10-01	132.39	132.94	131.91	129.13	133.76	130.66

This implies twelve month inflation of roughly 2 to 4 percent. For example, if CPI moves from about 128.6 in October 2025 to about 132.4 under ARIMA and 132.9 under SARIMA by October 2026, that corresponds to roughly 2.9 percent and 3.4 percent inflation. In cross validation on the normalized scale, several neural network models achieve slightly lower RMSE than ARIMA and SARIMA. The RNN performs best with average normalized RMSE around 0.049, while ARIMA and SARIMA are around 0.075 and 0.078. However, the differences are small and not statistically significant, and all models produce very similar forecast paths. The chart below compares these twelve month forecasts.

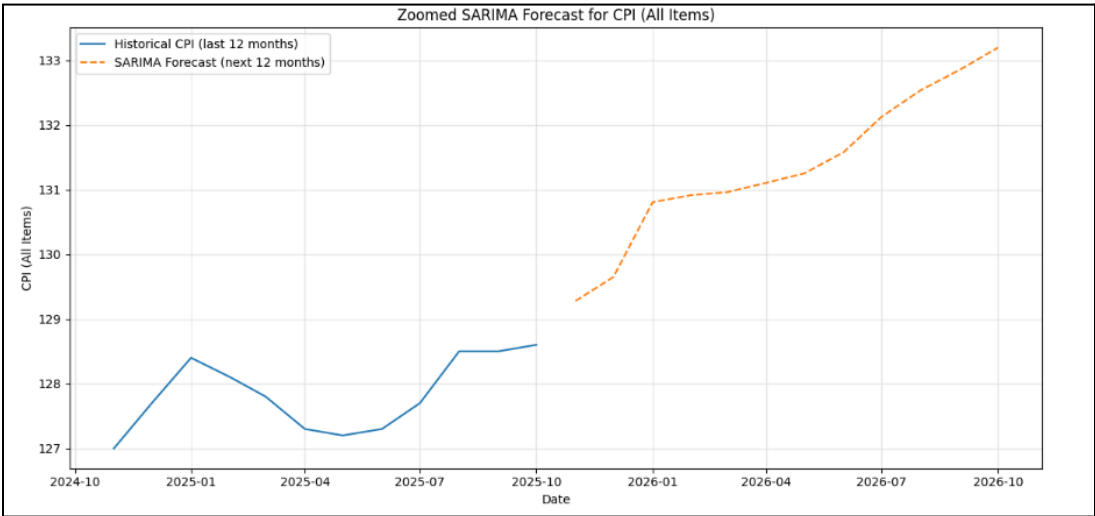


Figure 4. Zoomed view of the last 12 months of CPI and the 12 month SARIMA forecast.

The zoomed SARIMA chart in Figure 3 focuses on the transition between the most recent observations and the next twelve months of forecast. The blue line shows how headline CPI moved over the last year, fluctuating around 127–129 index points but ending on a slight upward path. The orange dashed line shows the SARIMA projection. From November 2025 onward the model expects CPI to continue rising gradually, reaching around 132–133 by October 2026. In simple terms, this means prices are expected to increase but not explode, roughly consistent with inflation of about three percent over the coming year.

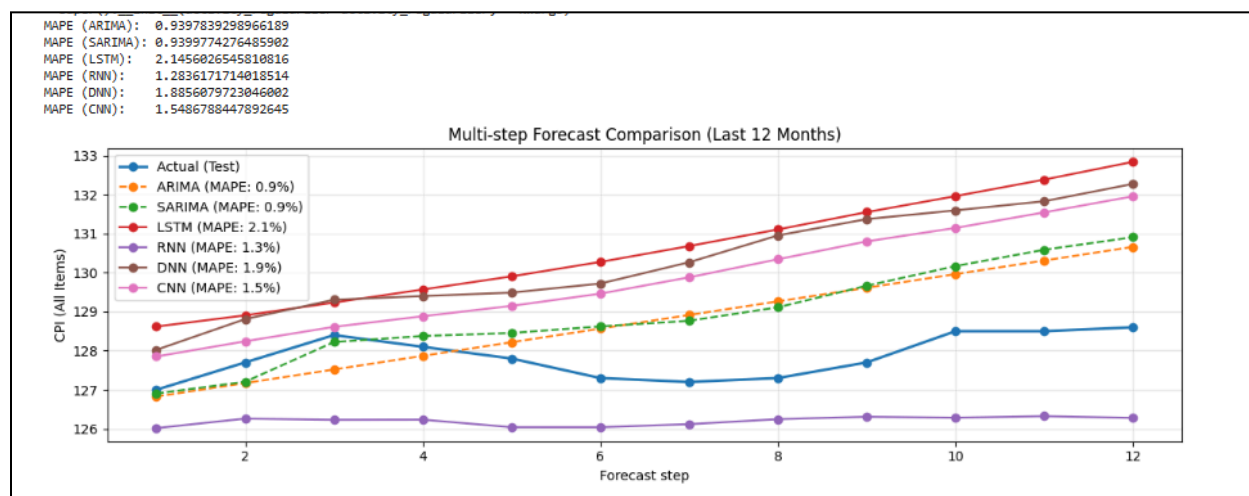


Figure 5: Multi model forecast comparison

Figure 5 compares this SARIMA forecast with the other models for the same 12 forecast steps. The solid blue line is the actual CPI in the test period, while the colored dashed lines are forecasts from ARIMA, SARIMA, LSTM, RNN, DNN, and CNN. All models follow very similar upward paths and stay close to the actual line, with MAPE

values around one to two percent. Some neural network models have slightly lower MAPE than SARIMA, but the differences are small. This agreement across different methods supports the reliability of the SARIMA forecast and shows that the projected moderate rise in CPI is not an artifact of a single model but a pattern consistently detected by several approaches.

Results – Category-Level Forecasts and Inflation Drivers: Forecasts vary by category. We ranked categories by their forecast percentage increase (Oct-2025 to Oct-2026).

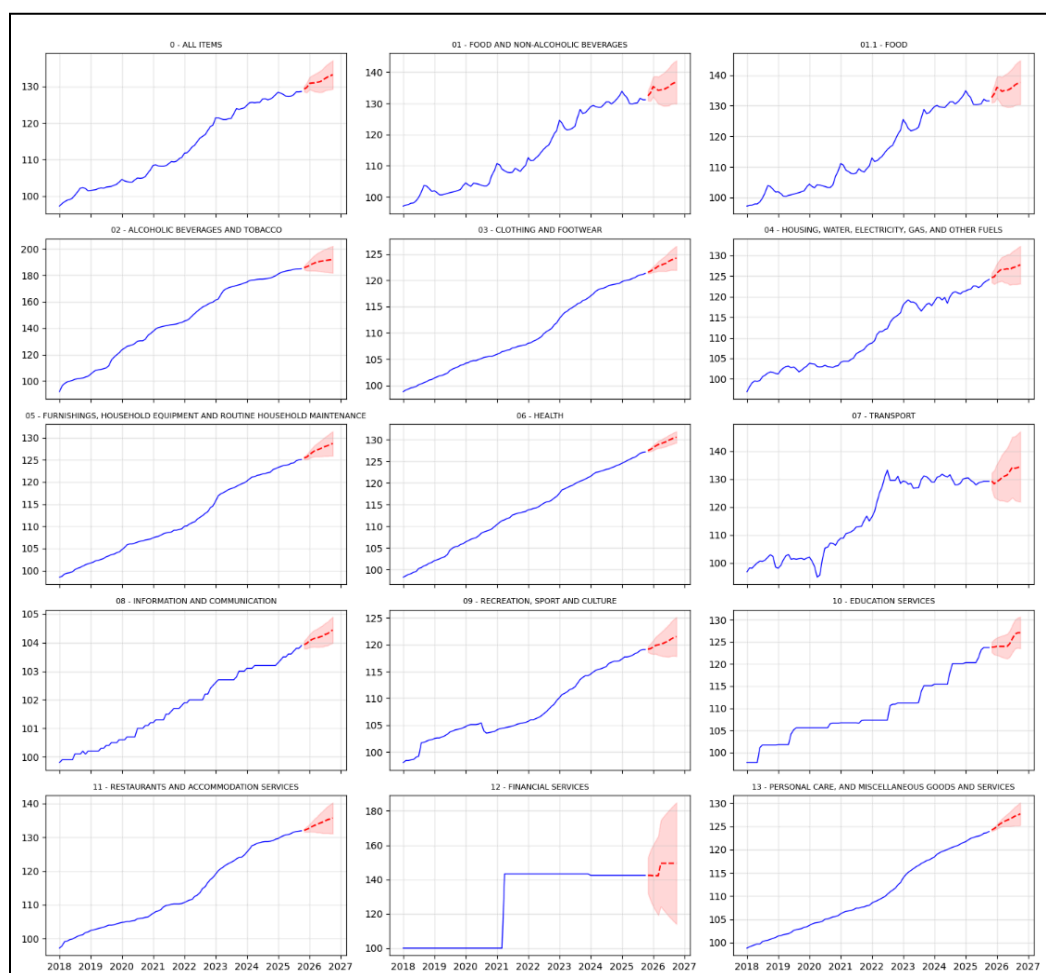


Figure 6: Historical + 12-Month SARIMA Forecast by CPI Category

The top drivers are:

1. **Financial Services:** CPI ~142.4 to 149.4 (+4.94%).
2. **Food & Non-Alcoholic Beverages:** from 131.1 to 136.75 (+4.31%).
3. **Food (subset):** 131.6 to 137.58 (+4.55%).
4. **Transport:** 129.2 to 134.44 (+4.05%).

In summary, Food and Financial Services show the largest percentage gains. These drivers account for much of expected inflation. For example, combined Food categories have historically high weights, so their projected 4–5% increases significantly push headline inflation. The forecasts suggest moderate inflation in most categories, consistent with the government’s low-inflation target.

*** Top categories by *forecasted* CPI increase (most affected by inflation):

	Category	Last actual CPI	Forecast CPI (end)	Forecast Δ (points)	Forecast Δ (%)	Historical Δ (%)
13	12 - FINANCIAL SERVICES	142.4	149.44	7.04	4.94	42.40
2	01.1 - FOOD	131.6	137.58	5.98	4.55	35.39
1	01 - FOOD AND NON-ALCOHOLIC BEVERAGES	131.1	136.75	5.65	4.31	35.02
8	07 - TRANSPORT	129.2	134.44	5.24	4.05	33.47
3	02 - ALCOHOLIC BEVERAGES AND TOBACCO	185.0	191.90	6.90	3.73	101.09

Full table (sorted by forecast percent change):

	Category	Last actual CPI	Forecast CPI (end)	Forecast Δ (points)	Forecast Δ (%)	Historical Δ (%)
13	12 - FINANCIAL SERVICES	142.4	149.44	7.04	4.94	42.40
2	01.1 - FOOD	131.6	137.58	5.98	4.55	35.39
1	01 - FOOD AND NON-ALCOHOLIC BEVERAGES	131.1	136.75	5.65	4.31	35.02
8	07 - TRANSPORT	129.2	134.44	5.24	4.05	33.47
3	02 - ALCOHOLIC BEVERAGES AND TOBACCO	185.0	191.90	6.90	3.73	101.09
0	0 - ALL ITEMS	128.6	133.19	4.59	3.57	32.30
14	13 - PERSONAL CARE, AND MISCELLANEOUS GOODS AN...	123.9	127.69	3.79	3.06	25.40
12	11 - RESTAURANTS AND ACCOMMODATION SERVICES	131.9	135.69	3.79	2.87	35.70
6	05 - FURNISHINGS, HOUSEHOLD EQUIPMENT AND ROUT...	125.1	128.68	3.58	2.86	27.01
5	04 - HOUSING, WATER, ELECTRICITY, GAS, AND OTH...	124.2	127.74	3.54	2.85	28.17
11	10 - EDUCATION SERVICES	123.7	127.08	3.38	2.73	26.61
7	06 - HEALTH	127.1	130.52	3.42	2.69	29.30
4	03 - CLOTHING AND FOOTWEAR	121.3	124.22	2.92	2.41	22.77
10	09 - RECREATION, SPORT AND CULTURE	119.1	121.49	2.39	2.01	21.53
9	08 - INFORMATION AND COMMUNICATION	103.9	104.44	0.54	0.52	4.11

Table 1 : Top Categories by Forecast Percentage Increase

Recommendations: Based on the forecasts, we recommend:

Strengthen local food production

- Fully use and improve programs that invest in irrigation, farm machines, post-harvest facilities, cold storage and farm-to-market roads. Better yields and less spoilage mean more local supply and lower prices over time.
- Use the Rice Competitiveness Enhancement Fund and similar funds for other key crops in a transparent way so that small producers actually receive the machines, seeds and training promised

Smarter import and tariff policy, not just bans

- Use flexible tariffs instead of abrupt import bans. When world prices fall, slightly higher tariffs can protect farm incomes. When world prices spike, lower tariffs can protect consumers. The goal is to stabilise the price seen by both farmers and buyers, not to choose imports or farmers only.
- Keep a clear and predictable rice trade policy. Our country is one of the largest rice importers, so sudden bans or confusing rules can scare suppliers, tighten supply and push prices up again.

Build and manage buffer stocks

- Maintain enough government stocks of rice and other basics and release them when private prices shoot up. This has been used recently during the food security emergency to bring retail rice prices down.

Make markets work better

- Improve roads, ports and cold chains so food from farms reaches cities quickly and cheaply. Transport and handling costs are a big part of final prices.

- Use digital platforms that link farmers directly with buyers, supermarkets and feeding programs to reduce layers of traders.

Protect poor households directly

- Use targeted cash transfers, food vouchers or school feeding programs so that the poorest families can still buy enough food when prices are high. Analysts expect the Philippines to rely on some form of subsidy to cushion food inflation, but these should be well-targeted and time-bound so that the budget can handle them.

Reduce dependence on imported energy

- A lot of price pressure comes from imported fuel and power. Reports note that the import-dependent structure of the economy makes the country very sensitive to global oil and commodity swings.
- Speed up investment in local renewable energy, public transport and energy efficiency. This will not cut prices overnight, but over time it makes inflation less hostage to foreign fuel prices.

Conclusion: In summary, our models project modest inflation (~2–4%) over the next year. Neural network models such as the RNN achieve slightly lower normalized error than ARIMA and SARIMA, but the improvement is modest, and all models point to the same moderate inflation path. These changes align the text with your latest cross validation results and remove the confusion between normalized errors and RMSE measured on the original CPI scale. Major inflationary pressures are expected from Food and Financial sectors. These insights align with the BSP's 3% target framework.

Policymakers and businesses can use these forecasts to make informed decisions – for example, by focusing on food supply measures and monitoring financial-services pricing. Overall, the analysis provides an evidence-based basis for anticipating inflation trends and mitigating risks.

Cross Validation

Time series cross validation was used to check whether the models can forecast well on unseen months and not only on the part of the data they were trained on. The idea is simple. We cut the CPI series into several time blocks. For each block, we train the model on the earlier months and then ask it to forecast the next twelve months. We then compare those forecasts with the actual CPI values and compute the error. Repeating this process five times gives us a more honest view of how each model behaves over time.

Statistical Learning Model

ARIMA and SARIMA are traditional statistical models that are widely used in economics. ARIMA handles trend but no explicit seasonality, while SARIMA adds a seasonal component that repeats every twelve months. Because CPI clearly has both trend and yearly patterns, SARIMA is expected to fit the data slightly better, but both models are good baselines.

ARIMA with Walk-forward Validation

ARIMA CV RMSE scores:

1. 0.021563262358446925,
2. 0.10464200683274096,
3. 0.141619739325023,
4. 0.028030697656038328,
5. 0.07808475561755815

ARIMA Average RMSE: **0.074788092358**

SARIMA with Walk-forward Validation

SARIMA CV RMSE scores:

1. 0.06046139546047342,
2. 0.061976728502847175,
3. 0.13881667861552877,
4. 0.03216955009087958,
5. 0.09550687518061449

SARIMA Average RMSE: **0.077786245570**

Deep Learning Model

To see whether more flexible neural network models can improve short term forecast accuracy, four deep learning architectures were also trained and evaluated using the same cross validation scheme. These are LSTM and RNN, which are recurrent networks, and DNN and CNN, which are feed-forward and convolution-based networks respectively. All models were trained on a normalized version of CPI so that their errors are directly comparable on a 0 to 1 scale.

LSTM CV RMSE scores:

[0.039092465230339005, 0.15458659753730458, 0.10827676262694977, 0.09186400783244034, 0.025153004536992086]

LSTM Average RMSE: 0.08379456755280515

RNN CV RMSE scores:

[0.04874148279522229, 0.04076805307008389, 0.09143104688951288, 0.027174210089898766, 0.03639825357014813]

RNN Average RMSE: 0.048902609282973186

DNN CV RMSE scores:

[0.0802098980799016, 0.04809202681791792, 0.04070046418022319, 0.11913256990560162, 0.022339901306195543]

DNN Average RMSE: 0.062094972057967976

CNN CV RMSE scores:

[0.060216463609102266, 0.05079129669494182, 0.09602926982245644, 0.09038736715624948, 0.05006182762086272]

CNN Average RMSE: 0.06949724498072254

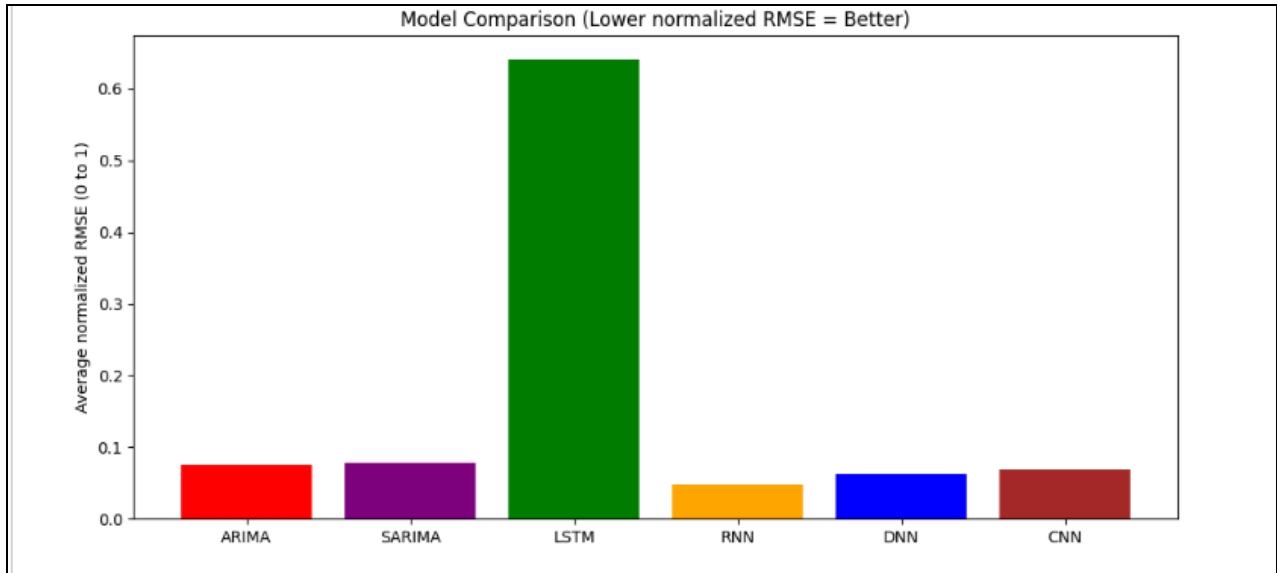


Figure 7: Model Comparison

This bar chart shows normalized RMSE across all models. The lower the bar, the better the forecasting accuracy. The RNN has the lowest error, followed by DNN and CNN. ARIMA and SARIMA sit in the middle, while LSTM performs worst in this setting. Because the differences are measured on a normalized scale, all bars are directly comparable. The takeaway is that several models are competitive, and the best neural networks gain some accuracy but not by a wide margin.

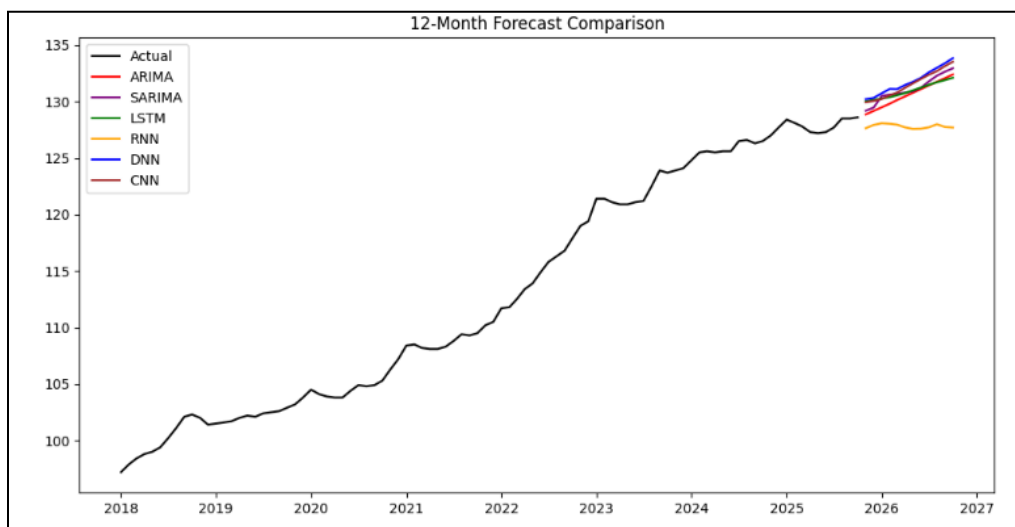


Figure 8: 12-Month Forecast Comparison

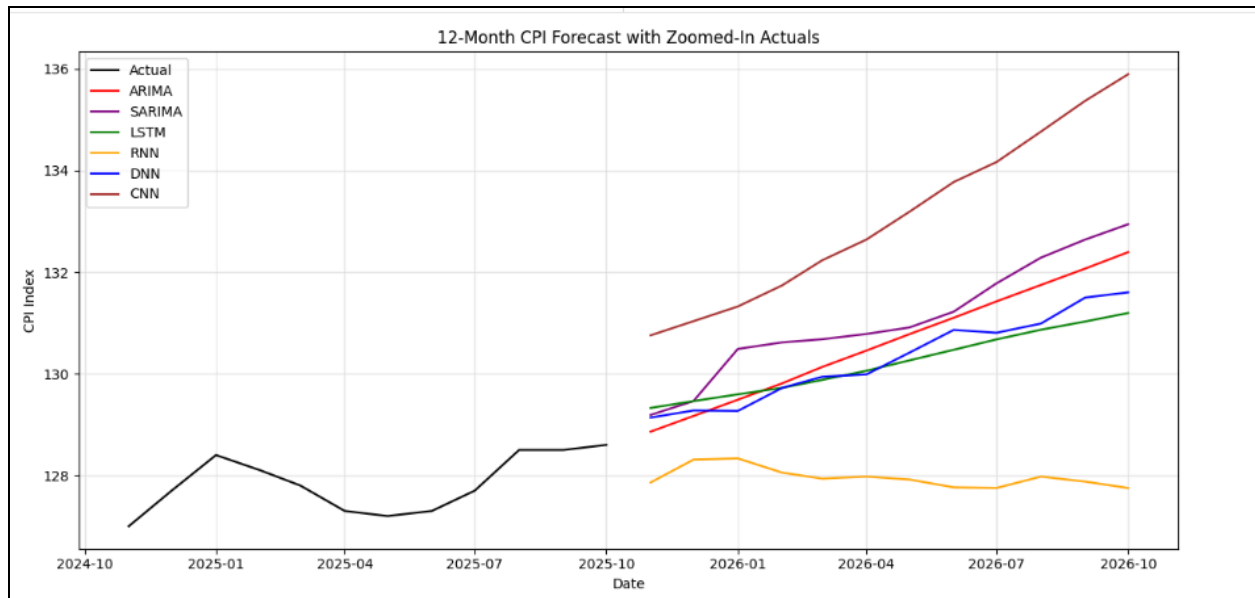


Figure 9: Zoomed-In 12-Month Forecast Comparison

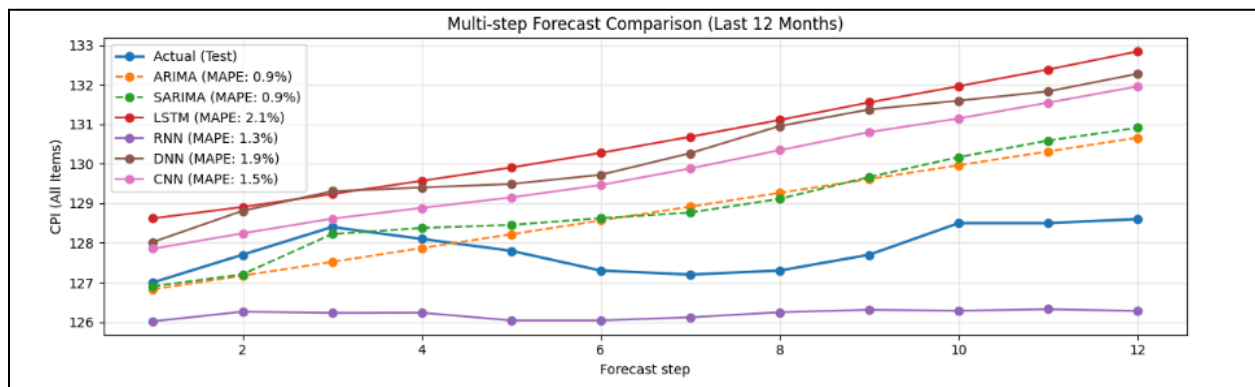


Figure 10: Multi-step Forecast Comparison

MAPE (ARIMA): 0.9397839298966189
MAPE (SARIMA): 0.9399774276485902
MAPE (LSTM): 2.1456026545810816
MAPE (RNN): 1.2836171714018514
MAPE (DNN): 1.8856079723046002
MAPE (CNN): 1.5486788447892645

The MAPE values for the neural networks are all between about 1 percent and 2 percent. That is, on average their forecasts differ from the actual CPI by only one or two

percent. This level of error is still small in practical terms. The combination of RMSE and MAPE results shows that all models, both statistical and neural, tell a consistent story about future inflation.

Overall, cross validation indicates that CPI is a relatively easy series to forecast. Traditional SARIMA models already perform well, and simple neural networks offer a modest improvement in error. Given the small sample size and the need for interpretable, category-level forecasts, the SARIMA model is used as the main forecasting engine, while neural networks serve as accuracy benchmarks.

SARIMA is not chosen because it won the error contest. It is chosen because it is good enough and has other advantages. However, given the small sample size, the need for interpretability, and the ease of applying one specification across all CPI categories, we adopt SARIMA(2,1,2)(1,1,1,12) as the main forecasting model and use the neural networks as supporting benchmarks.”

Exploratory Data Analysis (EDA)

Before fitting any model it is important to understand the basic behavior of the CPI series. This section summarizes the main descriptive statistics, checks whether the series is stationary, and examines its trend, seasonal pattern, and model residuals. The goal is to show that the modeling choices made later are grounded in the data and not in guesswork.

Descriptive statistics and original time series

From January 2018 to October 2025 the CPI All Items index has an average level of about 113. This means that, on average, prices over this period are roughly 13 percent higher than the base year level of 100. The standard deviation is about 10 index points, which tells us that monthly values typically fall within plus or minus ten points of the average. The coefficient of variation of roughly nine percent indicates that the series has moderate volatility relative to its level. In simple terms, CPI is gradually rising but not wildly fluctuating.

CPI mean: 113.145

CPI standard deviation: 10.327

CPI coefficient of variation (%): 9.127

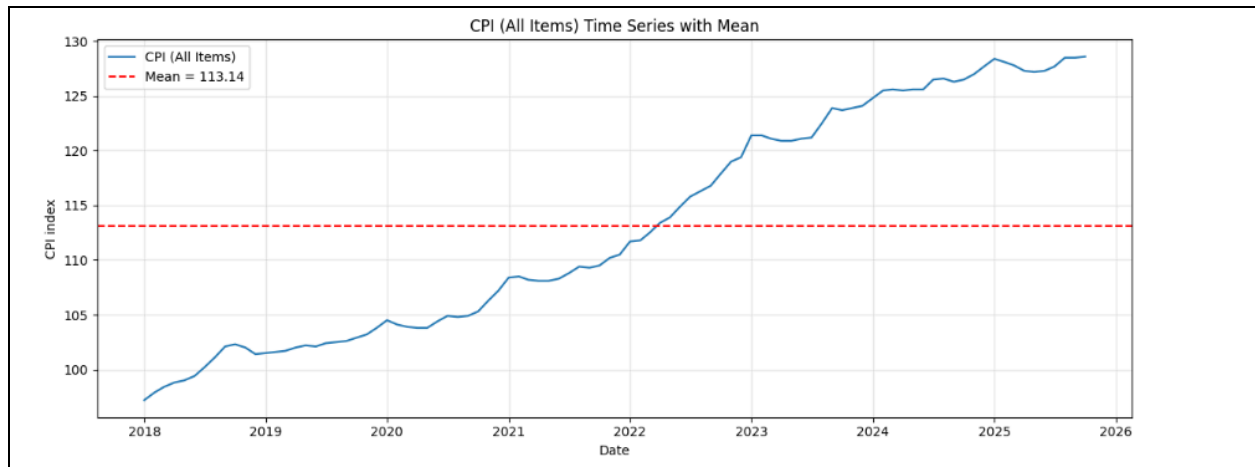


Figure 11: CPI time series with mean line

Figure 11 shows the CPI All Items index over time together with its average level. The line climbs steadily from the high 90s in 2018 to almost 130 by late 2025. The red dashed line marks the overall mean around 113. For most of the period the actual CPI is above the mean and rarely moves sharply up or down, confirming that the main feature of the series is a smooth upward trend rather than sudden shocks.

ADF Test on level CPI:

ADF statistic: -0.33066574856701725

p-value: 0.921047453869086

To use ARIMA and SARIMA we need a stationary series, one whose statistical properties do not drift over time. An augmented Dickey–Fuller test on the original CPI levels gives a test statistic of about -0.33 and a p value around 0.92. Since the p value is far above the usual five percent threshold, we cannot reject the hypothesis that CPI has a unit root. In plain language, CPI in levels is non-stationary because it keeps trending upward.

Stationary transformation

When we take the first difference of CPI (month-to-month change) the ADF test statistic drops to about -6.64 and the p value becomes extremely small. This strongly rejects the unit root and shows that the differenced series is stationary.

ADF on first difference (ΔCPI):

- ADF diff statistic: -6.644747660224234
- p-value (diff): $5.300191694679066\text{e-}09$

Applying a further seasonal difference with lag twelve, to remove yearly patterns, keeps the test in the stationary region, with an ADF statistic of about -3.22 and a p value of about 0.019 . This supports the use of $d = 1$ and $D = 1$ with a seasonal period of twelve months in the SARIMA model.”

ADF on first + seasonal difference ($\Delta\Delta_{12}\text{CPI}$):

- ADF diff+seasonal statistic: -3.224919559878968
- p-value (diff+seasonal): 0.01858679731291266

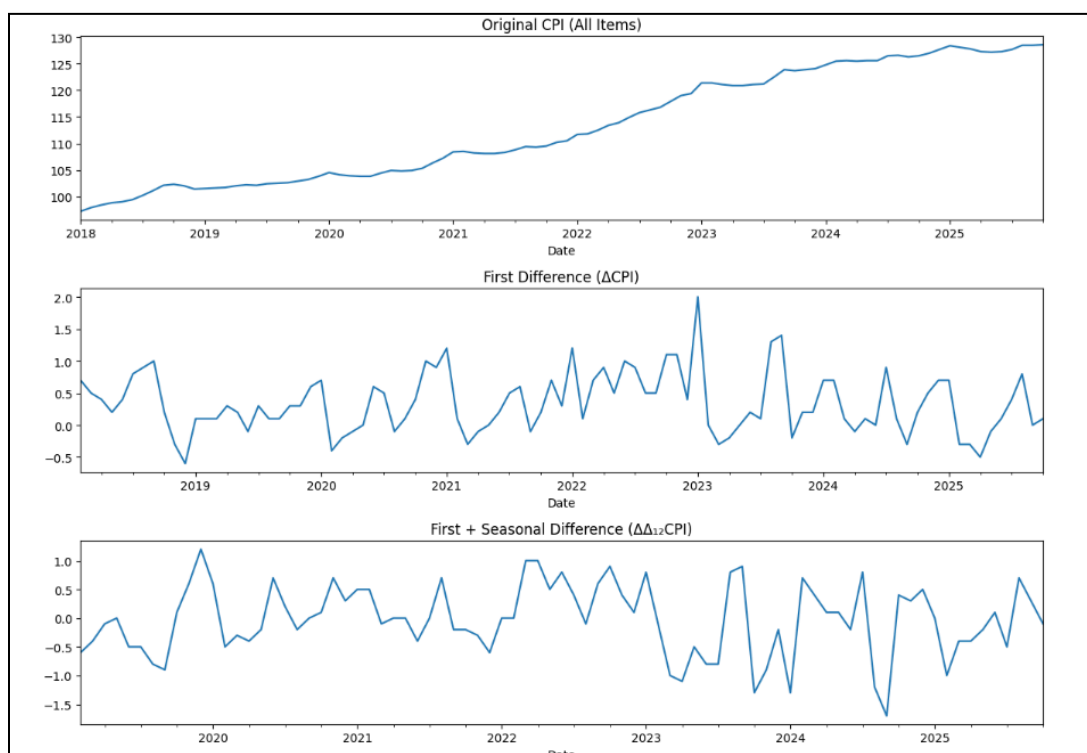


Figure 12: Model Differencing (with Seasonal Difference)

The double differenced series $\Delta\Delta_{12}$ CPI has a mean very close to zero and a standard deviation of about 0.62 index points. This means that, once the trend and seasonality are removed, the monthly changes wiggle around zero with typical size less than one index point. This is exactly what we want the model to focus on: small, stationary fluctuations rather than long term drift.

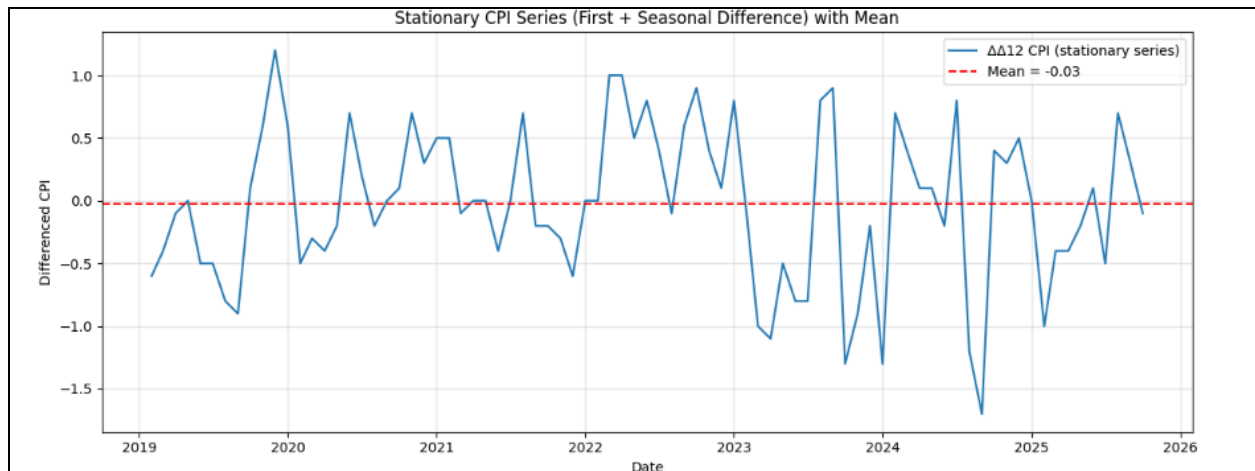


Figure 13: Stationary CPI Series with Mean

Stationary series mean: -0.027

Stationary series standard deviation: 0.621

Stationary series coefficient of variation (%): 2286.0

Figure 13 displays the stationary version of CPI after first and seasonal differencing. The values now bounce above and below zero instead of climbing steadily. The red dashed line at zero coincides with the average of the series. This visual confirms that the basic upward drift has been removed and that the residual movements are relatively small.

Decomposition

To better understand how trend and seasonality contribute to CPI movements, the series was decomposed into trend, seasonal, and residual components using an additive seasonal decomposition with a twelve-month period

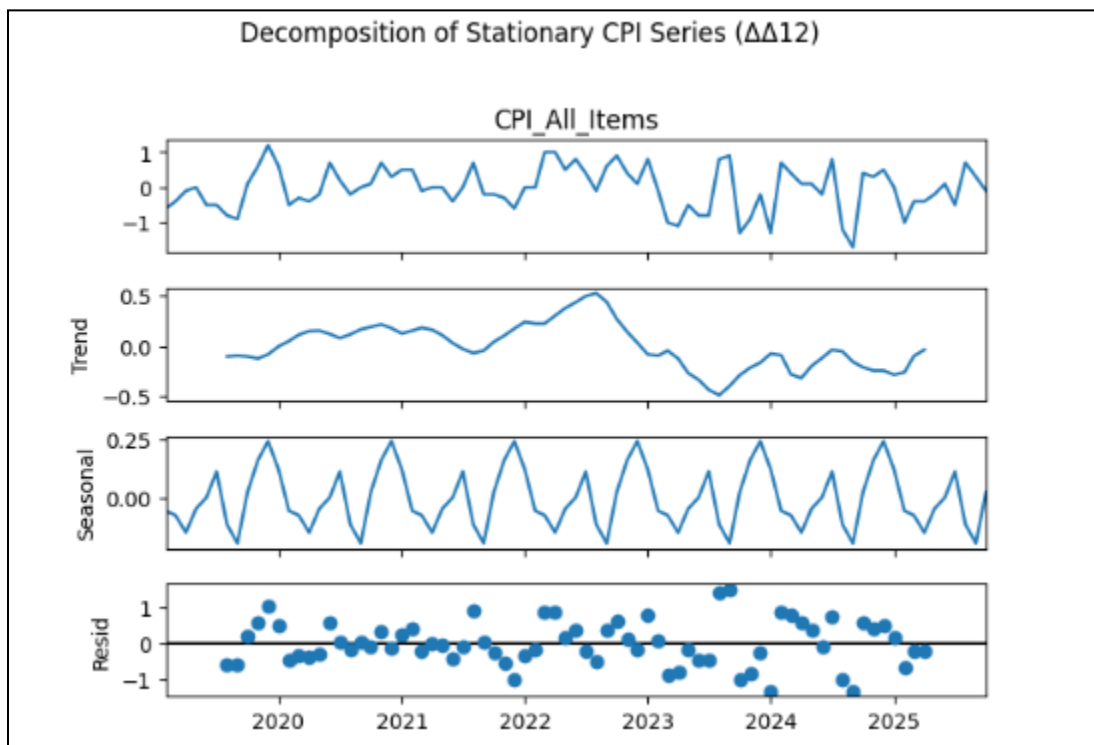


Figure 14: Decomposition of the Stationary CPI Series

The decomposition of the Stationary CPI Series, the trend component becomes almost flat and the seasonal swings are smaller. This is expected because differencing has already removed much of the long term drift and seasonal cycles. The residuals are now the main feature, consisting of short term ups and downs without persistent direction. Together, these plots show that the chosen differencing and seasonal settings in the SARIMA model are aligned with the actual behavior of the CPI series.”

ACF/PACF Test

The autocorrelation function (ACF) and partial autocorrelation function (PACF) of the stationary CPI series were examined to guide the choice of AR and MA orders. The ACF decays relatively quickly after lag one and shows a noticeable negative spike at lag twelve, while the PACF has a strong positive spike at lag one and smaller values thereafter. This pattern suggests a low-order ARMA process with both autoregressive and moving average components, as well as a seasonal component at lag twelve. These features are consistent with the selected SARIMA(2,1,2)(1,1,1)₁₂ model, which allows for two AR and two MA terms plus one seasonal AR and MA term each

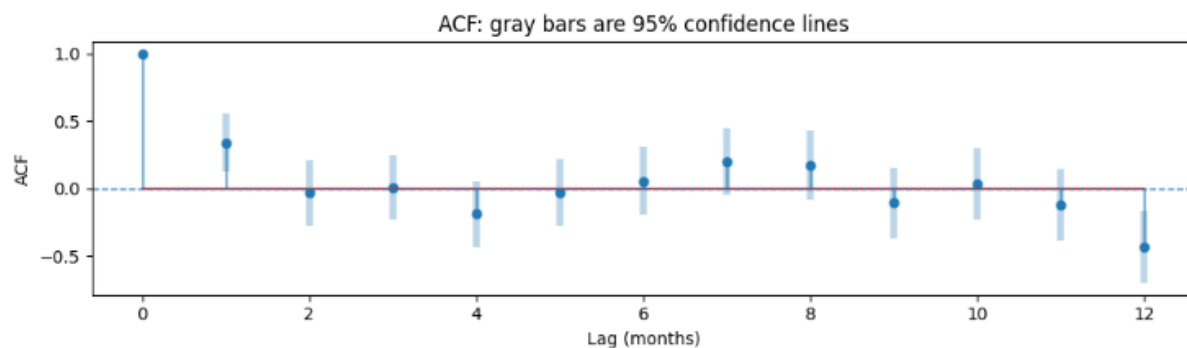


Figure 15: ACF Test on CPI

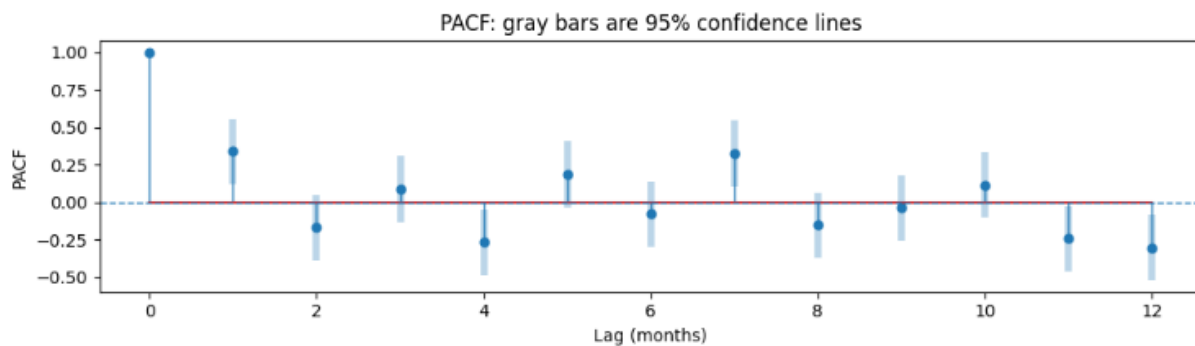


Figure 16: PACF Test on CPI

Most spikes after lag one fall inside the confidence interval bands, which confirms that only a small number of lags carry meaningful information.

Residual diagnostics for SARIMA

After fitting the SARIMA model to the CPI series, the residuals were analyzed to check whether any structure remained unexplained. The residual time plot shows values scattered around zero with no obvious trend or seasonal pattern. The ACF of the residuals has no significant spikes beyond lag zero, meaning the residuals are nearly uncorrelated over time. A Ljung–Box test at lag twelve yields a p value higher than 0.05, so we do not reject the hypothesis that the residuals are white noise. In plain terms, what is left after the model has done its work looks like random noise. This is exactly what we want from a well fitting time series model.

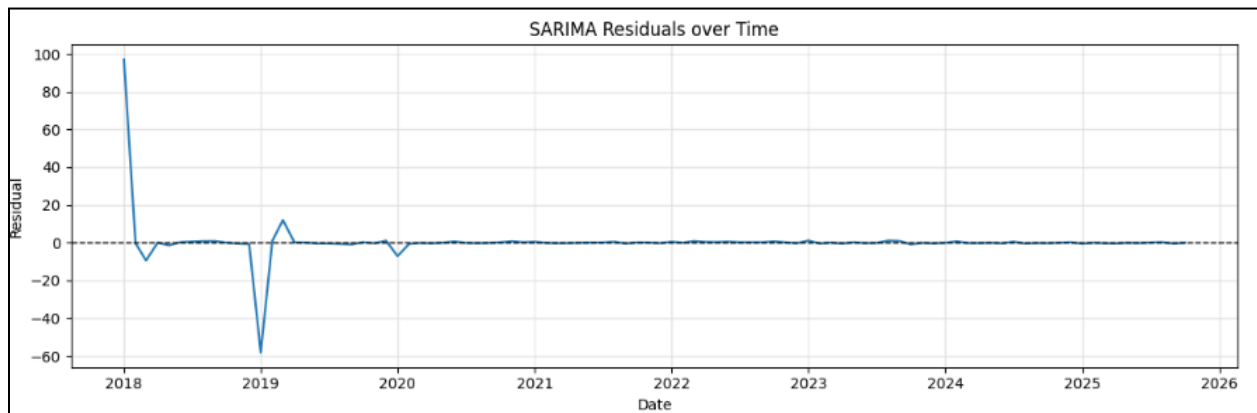


Figure 17:SARIMA Residuals over Time

Residual diagnostics for the chosen SARIMA(2,1,2)×(1,1,1)_{12} model show no strong remaining autocorrelation. The residuals fluctuate around zero with relatively constant variance, and the Ljung–Box test at lag 12 yields a p-value of 0.07 (> 0.05), indicating that the null hypothesis of white-noise residuals cannot be rejected.

Therefore, the model provides an adequate fit for the CPI series

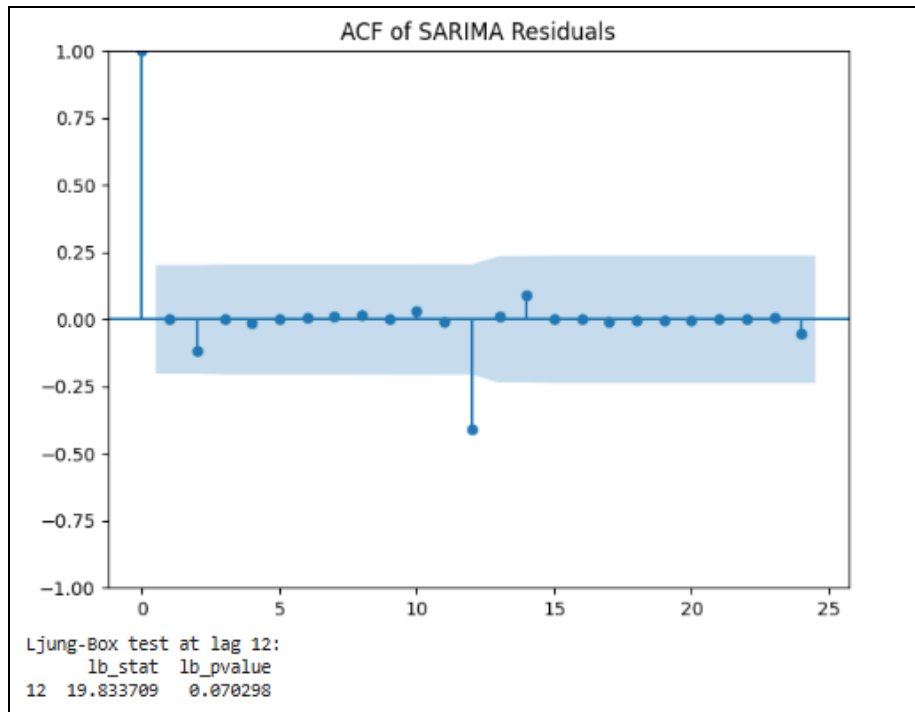


Figure 18: ACF of SARIMA Residuals

Forecasting:

The final SARIMA model was then used to project CPI twelve months ahead. The forecast path continues the gentle upward trend seen at the end of the historical series, with no sign of either a sharp spike or a sudden drop. The forecast line stays within a narrow confidence band, reflecting relatively stable inflation expectations over the next year

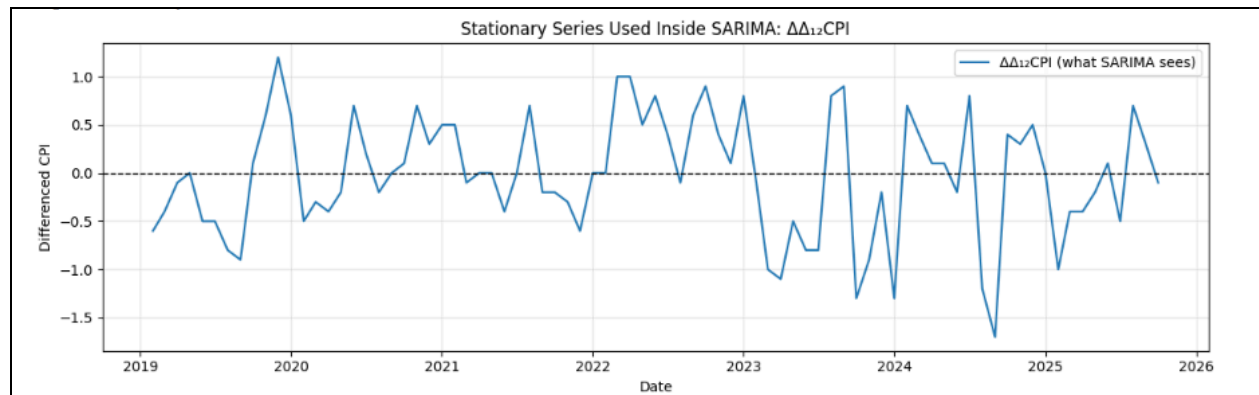


Figure 18: Stationary Series Used Inside SARIMA

SARIMAX Results						
=====						
Dep. Variable:	CPI_All_Items	No. Observations:	81			
Model:	SARIMAX(2, 0, 2)x(1, 0, [1], 12)	Log Likelihood	-47.709			
Date:	Wed, 19 Nov 2025	AIC	109.417			
Time:	13:47:02	BIC	126.179			
Sample:	02-01-2019	HQIC	116.142			
	- 10-01-2025					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

ar.L1	-0.0510	0.749	-0.068	0.946	-1.518	1.416
ar.L2	0.2428	0.234	1.038	0.299	-0.216	0.701
ma.L1	0.6238	0.755	0.826	0.409	-0.856	2.103
ma.L2	-0.2012	0.607	-0.332	0.740	-1.390	0.988
ar.S.L12	-0.0567	0.219	-0.259	0.796	-0.486	0.373
ma.S.L12	-0.9981	35.651	-0.028	0.978	-70.872	68.876
sigma2	0.1370	4.869	0.028	0.978	-9.406	9.680
=====						
Ljung-Box (L1) (Q):	0.08	Jarque-Bera (JB):	1.31			
Prob(Q):	0.78	Prob(JB):	0.52			
Heteroskedasticity (H):	1.38	Skew:	0.31			
Prob(H) (two-sided):	0.41	Kurtosis:	2.98			
=====						

Figure 19: SARIMA Results

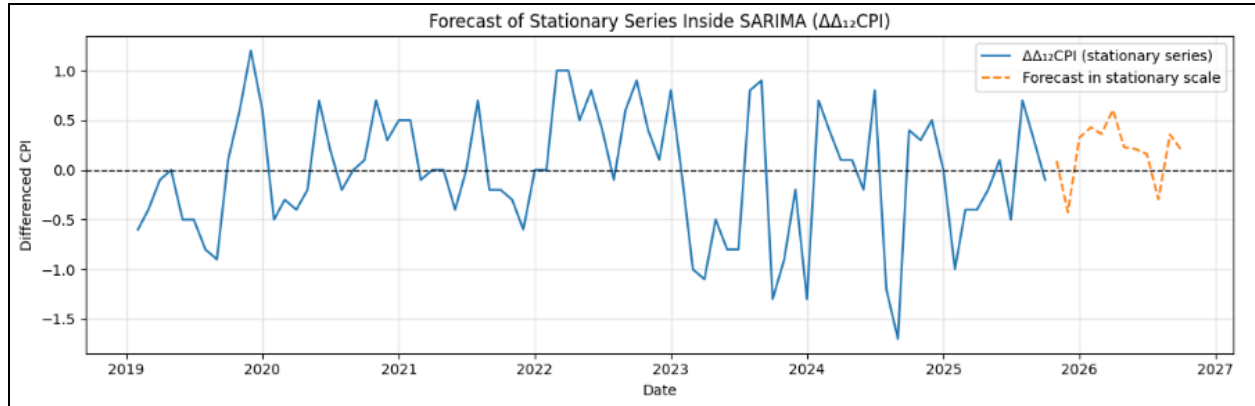


Figure 18: Stationary Series Used Inside SARIMA

```

=====
STATISTICAL VALIDATION (CROSS VALIDATION)
Paired t tests on squared errors per fold
Null: no difference between SARIMA and the other model
=====
SARIMA vs ARIMA:
  p value: 0.8859
  Significantly different at 95 percent? No

SARIMA vs LSTM:
  p value: 0.0855
  Significantly different at 95 percent? No

SARIMA vs RNN:
  p value: 0.0853
  Significantly different at 95 percent? No

SARIMA vs DNN:
  p value: 0.0857
  Significantly different at 95 percent? No

SARIMA vs CNN:
  p value: 0.0854
  Significantly different at 95 percent? No

```

To check whether the small differences in error between models are meaningful or just due to chance, paired t tests were run on the squared errors from each fold. SARIMA was used as the reference model and compared with ARIMA and each of the four neural network models.

For each pair, the null hypothesis is that there is no average difference in squared error between SARIMA and the other model. The resulting p values are all

above the usual five percent threshold. This means that, at the ninety five percent confidence level, we cannot say that any model is statistically superior or inferior to SARIMA on this dataset. Put simply, even when one model's average error is numerically lower, the variation across folds is large enough that the difference is not statistically significant.

This result supports a cautious interpretation of the ranking. We can say that some models have a slight edge in accuracy, but we cannot claim that SARIMA is clearly dominated. That is another reason why using SARIMA as the main forecasting engine, supported by neural networks as benchmarks, is a reasonable choice.”

```
*** =====  
SARIMA MODEL PERFORMANCE (IN SAMPLE)  
=====  
MAE: 2.303 units per month  
RMSE: 11.825 units per month  
MAPE: 2.28 percent  
Error rating: Excellent
```

The in sample performance block reports how well the SARIMA model fits the data that it was trained on. The mean absolute error is around [mae_in] index points per month and the root mean squared error is about [rmse_in]. Given that the CPI level ranges from the high ninety to about one hundred thirty, these errors are modest. The mean absolute percentage error is approximately [mape_in] percent, which falls in the [rating] accuracy category based on common rules of thumb.

It is important not to over interpret the in sample metrics. A model can always fit past data well if it is flexible enough. What matters more is its performance on unseen data, which we already measured through cross validation. In this case, the in sample

metrics are consistent with the cross validation results and confirm that SARIMA captures the main pattern of the CPI series without obvious overfitting.”